Advanced Probabilistic Machine Learning

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Graphical Models

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Overview

1. Graphical Models
2. Directed Graphical Models: Bayesian Networks
3. Undirected Graphical Models: Markov Random Fields

Overview

1. **Graphical Models**
2. Directed Graphical Models: Bayesian Networks
3. Undirected Graphical Models: Markov Random Fields

Graphical Models

- **Challenge**: How to formulate complex likelihoods/data models & priors for *actual* data?
  - **Example 1**: Match outcomes $y \in \{-1,1\}$ (data) for a head-to-head match between two players
    - **Prior**: $p(\boldsymbol{s}) = \mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2)$ skill belief
    - **Likelihood**: $p(y|\boldsymbol{s}) = \int \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)\ \mathrm{d}p_1\, \mathrm{d}p_2$ marginalization

      Match outcome

      Player performance
  - **Example 2**: Time series $\boldsymbol{y}$ of temperatures
    - **Prior**: $p(w) = \mathcal{N}(w; \mu, \sigma^2)$ External state mapping parameter belief
    - **Likelihood**: $p(\boldsymbol{y}|w, X) = \int \mathcal{N}(z_1;\ w \cdot x_1, \tau^2) \cdot \mathcal{N}(y_1; z_1, \beta^2) \cdot \mathcal{N}(z_2; z_1 + w \cdot x_2, \tau^2) \cdots\ \mathrm{d}\boldsymbol{z}$ marginalization

      Dynamics model

      Observed temperature model

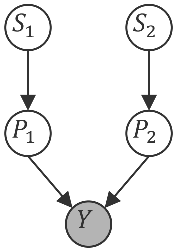
      Conditional hidden state model

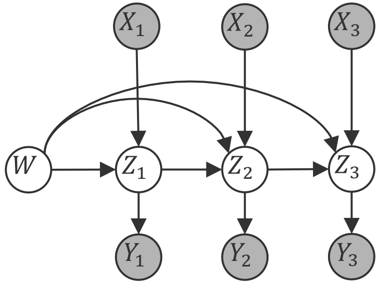
Graphical Models

- **Observation**: The product structure of the probabilities seems crucial
- **Idea**: Define a graph where each of the variables are nodes and edges indicate factor relationships between variables

$$\mathcal{N}(s_1;\mu_1,\sigma_1^2)\cdot\mathcal{N}(s_2;\mu_2,\sigma_2^2)\cdot\mathcal{N}(p_1;s_1,\beta^2)\cdot\mathcal{N}(p_2;s_2,\beta^2)\cdot\mathbb{I}(y(p_1-p_2)>0)$$

$$\mathcal{N}(w;\mu,\sigma^2)\cdot\mathcal{N}(z_1;\,w\cdot x_1,\tau^2)\cdot\mathcal{N}(y_1;z_1,\beta^2)\cdot\mathcal{N}(z_2;z_1+w\cdot x_2,\tau^2)\cdot\mathcal{N}(y_2;z_2,\beta^2)\cdots$$





- **Advantages**: Simple way to visualize factor structure of the joint probability
  - **Graphical Models**: Conditional independence based on graph properties
  - **Factor Graphs**: Efficient inference and approximation algorithms (next week)

Conditional Independence

- In modelling specific data, domain experts often know whether two (latent) measurements can affect each other or not (i.e., are independent)
  - **Examples**:
    - Skills of two players in a video game are not dependent if they never played before
    - Skills of two players in a video game *are* dependent if they have played many times!

- Graphical models are useful to determine conditional independence.
- **Conditional Independence**. *A random variable X_i is conditionally independent of a random variable X_j given the variable X_k if for all values X_k*

$$p\left(X_i = x_i | X_j = x_j, X_k = x_k\right) = p(X_i = x_i | X_k = x_k)$$

  - Equivalent definition: $p\left(X_i = x_i, X_j = x_j | X_k = x_k\right) = p(X_i = x_i | X_k = x_k) \cdot p\left(X_j = x_j | X_k = x_k\right)$
  - Shorthand notation (Dawid, 1979): $X_i \perp X_j | X_k$

Philip Dawid
(1946–)

Overview

1. Graphical Models
2. **Directed Graphical Models: Bayesian Networks**
3. Undirected Graphical Models: Markov Random Fields

Bayesian Networks

- **Observation**. Any joint distribution $p(x_1, \ldots, x_n)$ can be written as

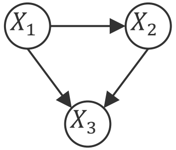
$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p(x_i | x_1, \ldots, x_{i-1})$$

- **Bayesian Network**. *Given a joint distribution*

$$p(x_1, \ldots, x_n) = \prod_{i=1}^{n} p\left(x_i | \boldsymbol{x}_{\text{parents}_i}\right)$$

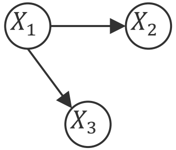
a Bayesian network *is a graph with a node for every variable X_i, and a directed edge from every variable $X_j, j \in \text{parent}_i$* to X_i.

- **Examples**: For 3 variables, we have these four generic Bayesian networks



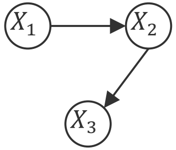
$p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1, x_2)$

full mesh



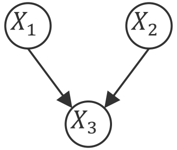
$p(x_1, x_2, x_3) = p(x_1) \cdot \prod_{i=2}^{3} p(x_i|x_1)$

star



$p(x_1, x_2, x_3) = p(x_1) \cdot \prod_{i=2}^{3} p(x_i|x_{i-1})$

chain



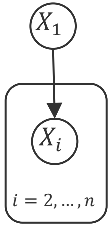
$p(x_1, x_2, x_3) = p(x_3|x_1, x_2) \cdot \prod_{i=1}^{2} p(x_i)$

sink

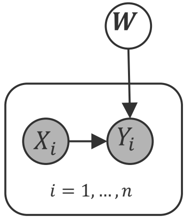
Bayesian Network Models

- **Plate**. If a subset of variables has the same relation only differing in their index, we use a ”plate” to collapse them into a single graphical element.
  - Increase readability of models for large amounts of parameters and data
- A Bayesian network must always be a **directed acyclic graph** because only those have a topological order corresponding to a variable order.
- **Observed Variables**. If a subset of variables has been observed (“data”), the variable nodes are usually shaded (“clamped”).
  - **Example**: Discriminatory Models

$$p(x_1, x_2, \dots, x_n) = p(x_1) \cdot \prod_{i=2}^{n} p(x_i|x_1)$$



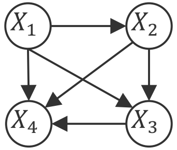
$$p(\boldsymbol{w}, y_1, \dots, y_n | x_1, \dots, x_n) = \prod_{i=1}^{n} p(y_i | x_i, \boldsymbol{w}) \cdot p(\boldsymbol{w})$$



$$p\big(\boldsymbol{w} | (x_1, y_1), \dots, (x_n, y_n)\big) \propto \prod_{i=1}^{n} p(y_i | x_i, \boldsymbol{w}) \cdot p(\boldsymbol{w})$$

Representation Complexity

- For simplicity, let us assume that $x_i \in \{1, \dots, K\}$



Naive

$p(x_1, \dots, x_n)$

x_1	x_2	x_3	x_4	$p(x_1, x_2, x_3, x_4)$
1	1	1	1	p_{1111}
1	1	1	2	p_{1112}
		$\vdots$		
1	1	1	K	p_{111K}
1	1	2	1	p_{1121}
		$\vdots$		
K	K	K	K	$1-\sum$

$K^4 - 1$

Bayesian Network

$p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1, x_2) \cdot p(x_4|x_1, x_2, x_3)$

x_1	$p(x_1)$
1	p_1
2	p_2
$\vdots$	
K	$1-\sum$

$K - 1$

x_2	x_1	$p(x_2|x_1)$
1	1	p_{11}
2	1	p_{21}
	$\vdots$	
K	1	$1-\sum$
1	2	p_{12}
	$\vdots$	
K	K	$1-\sum$

$(K-1) \cdot K$

x_3	x_1	x_2	$p(x_3|x_1, x_2)$
1	1	1	p_{111}
2	1	1	p_{211}
	$\vdots$		
K	1	1	$1-\sum$
1	1	2	p_{112}
	$\vdots$		
K	K	K	$1-\sum$

$(K-1) \cdot K^2$

x_4	x_1	x_2	x_3	$p(x_4|x_1, x_2, x_3)$
1	1	1	1	p_{1111}
2	1	1	1	p_{2111}
		$\vdots$		
K	1	1	1	$1-\sum$
1	1	1	2	p_{1112}
		$\vdots$		
K	K	K	K	$1-\sum$

$(K-1) \cdot K^3$

$$(K-1) \cdot (1 + K + K^2 + K^3) = (K + K^2 + K^3 + K^4) - (1 + K + K^2 + K^3)$$
$$= K^4 - 1$$

Sampling a Bayesian Network

- One advantage of a Bayesian network is the ability to *sample* $p(x_1, \ldots, x_n)$

Ancestral Sampling

1. Topologically sort all variables $X_1, \ldots, X_n$ into $X_{(1)}, \ldots, X_{(n)}$
2. Sample each variable $X_{(i)}$ using distribution $p\left(X_{(i)} | X_{(1)}, \ldots, X_{(i-1)}\right)$

- **Assumption**
  1. Sampling from the conditional distributions is simpler than from the joint distribution
  2. There are no clamped nodes, that is, we do not condition on any variable
- **Problems**
  1. Sampling is *sequential* one variable at the time
  2. Conditioning happens only on frequent events because

$$p(x_1, \ldots, x_{n-1} | x_n) = \frac{p(x_1, \ldots, x_n)}{p(x_n)} \approx \frac{\left|\left\{\left(x_{j,1}, \ldots, x_{j,n}\right) \mid x_{j,1} = x_1 \wedge \cdots \wedge x_{j,n} = x_n\right\}\right|}{\boxed{\left|\left\{x_{j,n} = x_n\right\}\right|}}$$

≈ 0 for rare events

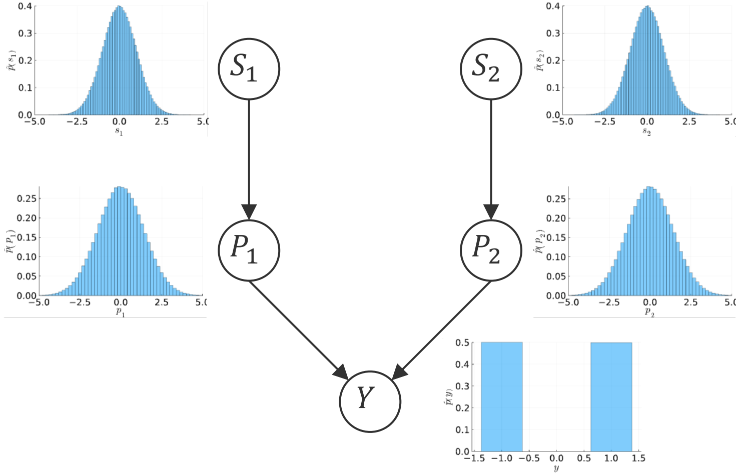
Sampling a Bayesian Network: Example

```
# samples from the TrueSkill graphical model
function sample(; n = 100000, μ1=0.0, σ1=1.0, μ2=0.0, σ2=1.0, β=1.0)
    samples = Vector{Vector{Float64}}(undef, n)
    for i in 1:n
        s1 = rand(Normal(μ1, σ1))
        s2 = rand(Normal(μ2, σ2))
        p1 = rand(Normal(s1, β))
        p2 = rand(Normal(s2, β))
        y = p1 > p2 ? 1.0 : -1.0
        samples[i] = [s1, s2, p1, p2, y]
    end
    return samples
end
```

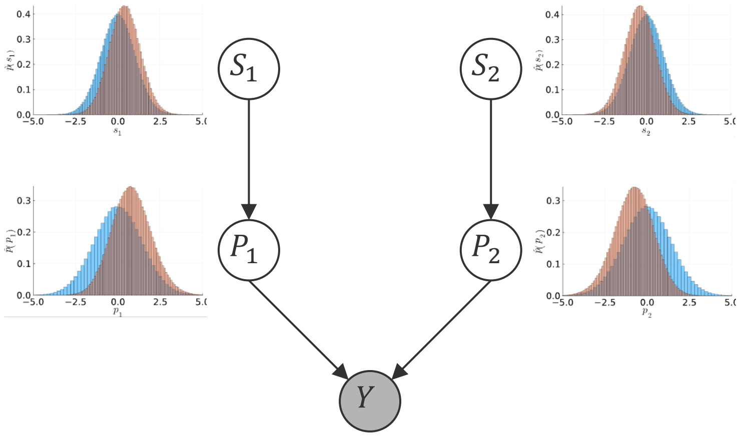
$$\boxed{\mathcal{N}(s_1;\mu_1,\sigma_1^2)\cdot\mathcal{N}(s_2;\mu_2,\sigma_2^2)\cdot\mathcal{N}(p_1;s_1,\beta^2)\cdot\mathcal{N}(p_2;s_2,\beta^2)\cdot\mathbb{I}(y(p_1-p_2)>0)}$$

$\mu_1=\mu_2$

Without match outcome



With match outcome ($y=1$)

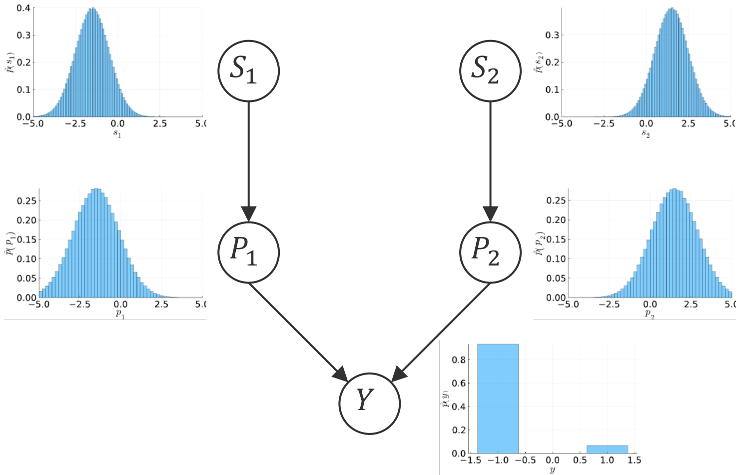


Sampling a Bayesian Network: Example (ctd)

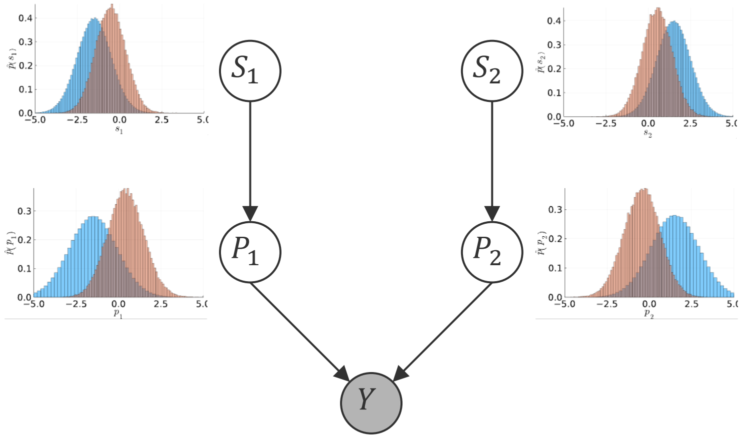
$$\boxed{\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)}$$

$\mu_1 \ll \mu_2$

Without match outcome



With match outcome ($y = 1$)



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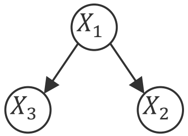
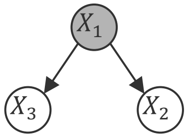
Unit 3 – Graphical Models

Conditional Independence: Warm-Up I

- **Tail-to-Tail Node (x_1):** $p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1)$





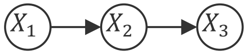
$$p(x_2, x_3) = \sum_{\{x_1\}} p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1) \neq p(x_2) \cdot p(x_3)$$

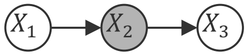
not independent

$$p(x_2, x_3|x_1) = \frac{p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_1)}{p(x_1)} = p(x_2|x_1) \cdot p(x_3|x_1)$$

conditionally independent

- **Head-to-Tail Node (x_2):** $p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2|x_1) \cdot p(x_3|x_2)$





$$p(x_1, x_3) = p(x_1) \cdot \sum_{\{x_2\}} p(x_2|x_1) \cdot p(x_3|x_2) = p(x_1) \cdot p(x_3|x_1) \neq p(x_1) \cdot p(x_3)$$

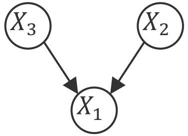
not independent

$$p(x_1, x_3|x_2) = \frac{p(x_2|x_1) \cdot p(x_1)}{p(x_2)} \cdot p(x_3|x_2) = p(x_1|x_2) \cdot p(x_3|x_2)$$

conditionally independent

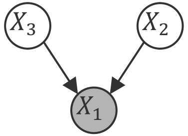
Conditional Independence: Warm-Up II

- **Head-to-Head Node (x_1):** $p(x_1, x_2, x_3) = p(x_2) \cdot p(x_3) \cdot p(x_1|x_2, x_3)$



$$p(x_2, x_3) = \sum_{\{x_1\}} p(x_1|x_2, x_3) \cdot p(x_2) \cdot p(x_3) = p(x_2) \cdot p(x_3)$$

independent

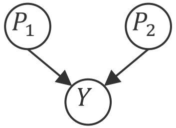


$$p(x_2, x_3|x_1) = \frac{p(x_1|x_2, x_3) \cdot p(x_2) \cdot p(x_3)}{p(x_1)} \neq p(x_2|x_1) \cdot p(x_3|x_1)$$

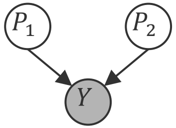
not (always) conditionally independent

  - It can be shown that the path between X_2 and X_3 are only independent if *none* of the *descendant* node from X_1 (that can be reached in the directed graph) is observed!

- **Skill Example (ctd)**: Consider the performance of two players



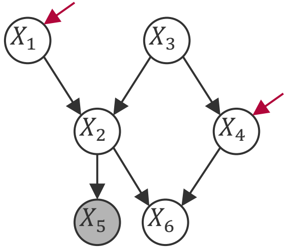
Before match: P_1 and P_2 are independent



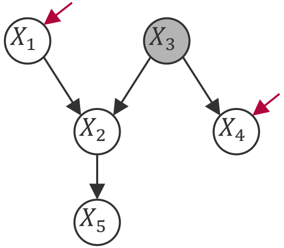
After match: P_1 and P_2 are **not** independent

Conditional Independence: d-separation

- **Blocked Node**. *A node in a Bayesian network is said to be blocked if*
  1. *It's a head-to-tail or tail-to-tail node and the node is observed.*
  2. *It's a head-to-head node and neither the node nor any of its descendants are observed.*
- **d-separation**. *Given a Bayesian network and a subset of observed variables, two non-observed variables X_i and X_j are* conditionally independent *(that is, d-separated) if every undirected path between X_i and X_j contains at least one blocked node.*
- **Examples**.



X_1 and X_4 are not independent



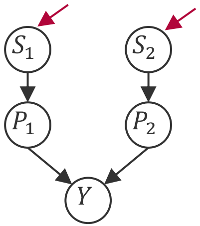
X_1 and X_4 are independent

Judea Pearl
(1936–)

Conditional Independence: Skill Example

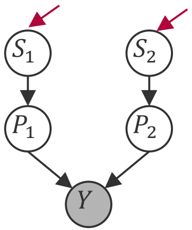
- **Skill Example (ctd)**: Consider the skills of two players

Before match



S_1 and S_2 are independent

After match



S_1 and S_2 are **not** independent

- Intuitive because
  - Before the match there is no information that "links" the skill of two players
  - After the match, if the skill of the winning player goes down (e.g., due to a loss in a subsequent match) then the skill of the opponent also needs to go down (or otherwise the observed match outcome would not have been possible)

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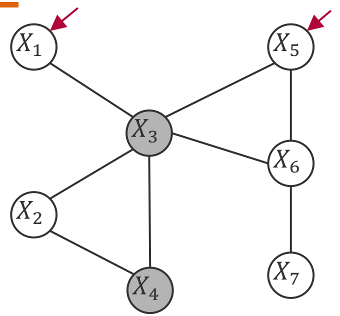
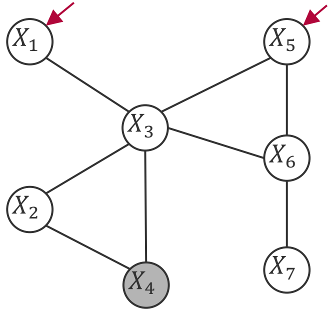
Conditional Independence Revisited

- **Observation**: Bayesian networks are visual representations of conditional distributions, but conditional independence requires checking a graph property (d-separation) that is
  - based on removing the direction of the arrows
  - not entirely local
- **Idea**: Remove the direction of arrows in a graphical model and define conditional independence by pure connectivity!
- **Conditional Independence** . *Given an undirected graphical model and a subset D of observed variables, two variables X_i and X_j are* conditionally independent *if there is no path between X_i and X_j when removing all variables $X_k \in D$.*
  - This property is also called the *Markov property* of the graph of random variables X_i
  - **Pro**: Very easy to verify conditional independence of variable sets
  - **Con**: Given this conditional independence definition, what is the functional form of the joint probability that satisfies this condition?



X_1 and X_5 are conditionally independent



X_1 and X_5 are **not** conditionally independent

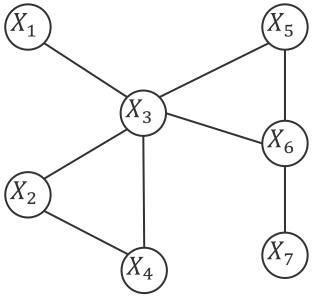
Hammersley-Clifford Theorem

- **Hammersley-Clifford (1971)**. Given n random variables $X_1, \ldots, X_n$, a density p satisfies the Markov property for an undirected graph G over $X_1, \ldots, X_n$ if and only

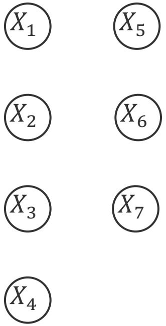
$$p(\boldsymbol{x}) \propto \prod_{C \in \mathrm{cliques}(G)} \psi_C(\boldsymbol{x}_C)$$

where $\psi_C(\cdot)$ are strictly positive functions and $\boldsymbol{x}_C$ are the values of $\boldsymbol{x}$ indexed by C.

  - Computing the density up to normalization is computationally easy but the normalization constant is computationally hard as it requires sums/integrals of products of potentials!



Cliques of size 1



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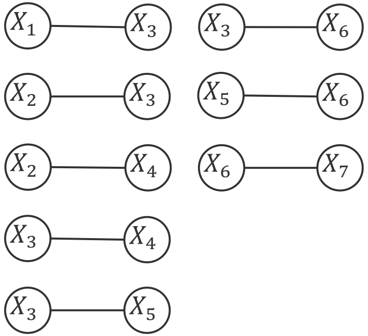
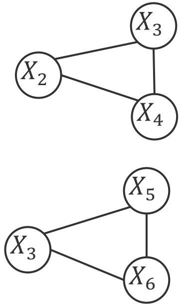


Cliques of size 3



John Hammersley (1920 – 2004)

Peter Clifford (1943 – 2025)

Proof of the Hammersley-Clifford Theorem

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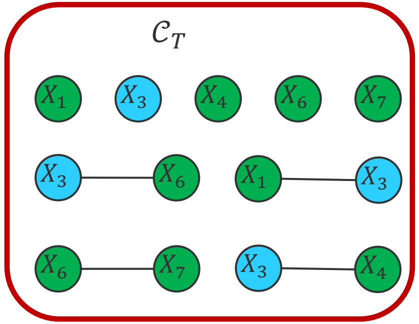
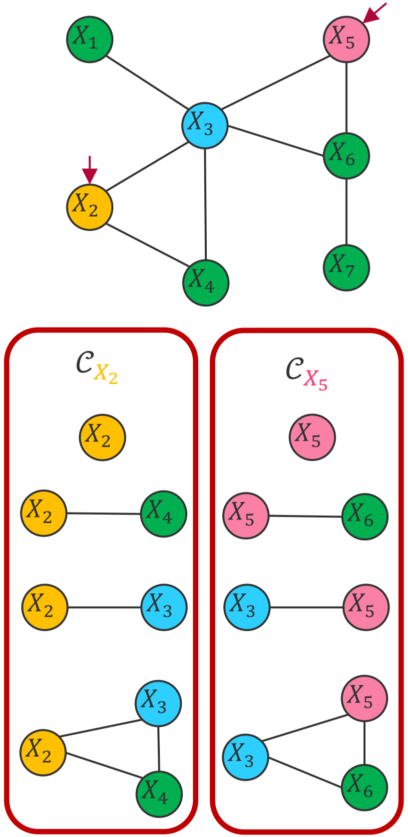
- Assume that $p(\boldsymbol{x}) \propto \prod_{C \in \mathrm{cliques}(G)} \psi_C(\boldsymbol{x}_C)$ and pick X_i, X_j and D such that removing all variables in D from the graph G separates X_i, X_j. Define $S = V \setminus \left(D \cup \{X_i\} \cup \{X_j\}\right)$
- Then $\mathrm{cliques}(G)$ separates in three disjoint sets

$$\mathcal{C}_{X_i} := \left\{C : C \cap \{X_i\} \neq \emptyset \wedge C \cap \{X_j\} = \emptyset\right\}$$
$$\mathcal{C}_{X_j} := \left\{C : C \cap \{X_i\} = \emptyset \wedge C \cap \{X_j\} \neq \emptyset\right\}$$
$$\mathcal{C}_T := \left\{C : C \cap \{X_i\} = \emptyset \wedge C \cap \{X_j\} = \emptyset\right\}$$





- Thus, the joint density has the following property

$$p(\boldsymbol{x}) \propto \left[\prod_{C \in \mathcal{C}_{X_i}} \psi_C(\boldsymbol{x}_C)\right] \cdot \left[\prod_{C \in \mathcal{C}_{X_j}} \psi_C(\boldsymbol{x}_C)\right] \cdot \left[\prod_{C \in \mathcal{C}_T} \psi_C(\boldsymbol{x}_C)\right]$$

- Because all variables in D separate X_i, X_j we have $S = S_i \cup S_j \cup S_O$ which are disjoint where S_i are the variables in S which are in cliques in $\mathcal{C}_{X_i}$ (and similar for S_j). Thus

$$p(x_i, x_j | \boldsymbol{x}_D) \propto \underbrace{\left[\sum_{\{\boldsymbol{x}_{S_i}\}} \prod_{C \in \mathcal{C}_{X_i} \cup \mathcal{C}_T} \psi_C(\boldsymbol{x}_C)\right]}_{q_i(x_i)} \cdot \underbrace{\left[\sum_{\{\boldsymbol{x}_{S_j}\}} \prod_{C \in \mathcal{C}_{X_j} \cup \mathcal{C}_T} \psi_C(\boldsymbol{x}_C)\right]}_{q_j(x_j)} \cdot \underbrace{\left[\sum_{\{\boldsymbol{x}_S\}} \prod_{C \in \mathcal{C}_T} \psi_C(\boldsymbol{x}_C)\right]}_{\text{constant}}$$

Sampling an Undirected Graphical Model

- If the X_i are discrete random variables, we can *sample* $p(x_1, \dots, x_n)$ as follows

Gibbs Sampling

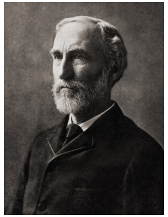
1. Initialize all random variables with a value $x_1, \dots, x_n$
2. For all $i \in \{1, \dots, n\}$, $x_i \sim p(X_i | X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1} \dots, X_n = x_n)$
3. Skip K such generated samples $\boldsymbol{x}$ to remove

- **Assumption**: There are only a few cliques $\mathcal{C}_{x_i} \coloneqq \{C : C \cap \{X_i\} \neq \emptyset\}$ because

$$p(X_i | X_1 = x_1, \dots, X_{i-1} = x_{i-1}, X_{i+1} = x_{i+1} \dots, X_n = x_n) \propto \prod_{C \in \mathcal{C}_{X_i}} \psi_C(\boldsymbol{x}_C)$$

- **Problem**: Step 2 produced highly correlated samples (thus we skip K samples $\boldsymbol{x}$)

- **Advantage**: We can easily condition on observed variables thereby reducing the number of elements of $\boldsymbol{x}$ in step 2!

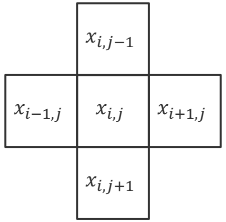


Josiah W. Gibbs
(1839 – 1903)

Donald Geman
(1943 –)

Sampling an Undirected Graphical Model

- **Ising Model**: Defined on an $L \times L$ grid of binary labels $x_{i,j} \in \{-1, +1\}$ with 4-neighbourhood $\mathfrak{N}(i,j) = \{(i+1,j), (i-1,j), (i,j+1), (i,j-1)\}$



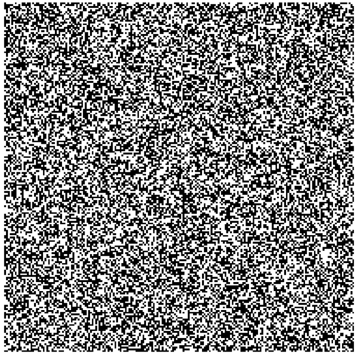
Coupling strength that a pixel has the same label as its neighbours

$$p(\boldsymbol{x}) \propto \exp\left(\sum_{i=1}^{L}\sum_{j=1}^{L}\left[\boxed{h} \cdot x_{i,j} + \sum_{(i',j') \in \mathfrak{N}(i,j)} \boxed{J} \cdot x_{i,j} \cdot x_{i',j'}\right]\right)$$

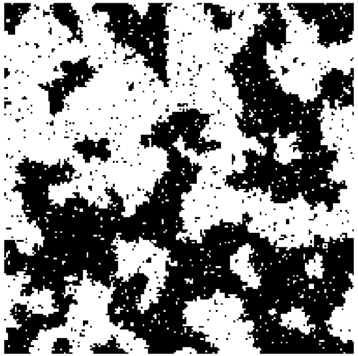
With zero coupling, $h = \frac{1}{2} \cdot \log\left(\frac{p(x_{i,j}=+1)}{1-p(x_{i,j}=+1)}\right)$

```julia
# update a single bit using Gibbs sampling
function gibbs_update!(model::IsingModel, i::Int, j::Int)
    L, h, J = model.L, model.h, model.J

    # Calculate the local field
    local_field = h + J * (
        model.x[mod1(i+1, L), j] +
        model.x[mod1(i-1, L), j] +
        model.x[i, mod1(j+1, L)] +
        model.x[i, mod1(j-1, L)]
    )

    # Compute the probability of a state being +1
    p_plus = 1 / (1 + exp(-2 * local_field))

    # Update the single bit
    model.x[i, j] = rand() < p_plus ? 1 : -1

    return model
end
```

$J = 0, h = 0$



$J = 0.5, h = 0$



$J = 1.0, h = 0$

Summary

- **Graphical Models**
  - Simple way to visualize the product structure of a joint probability distribution
  - Useful for modelling real-life data generating processes
  - Allows to test for conditional independence
- **Bayesian Networks**
  - A directed acyclic graph where each edge points from a conditioning to a conditioned variable in the model (often easier to formulate for experts)
  - A generative model of the data that can be easily sampled from (ancestral sampling)
  - d-separation is a set of simple rules ("blocking") to read off conditional independence
- **Markov Random Field**
  - An undirected graph where the probability is proportional to a product of positive potential functions over all cliques
  - A generative model of the data that can be easily sampled from (Gibbs sampling)
  - Connectivity is the graph property to read off conditional independence

See you next week!